

The Trading Ecosystem

From discovery to evidence, authority, execution and reevaluation

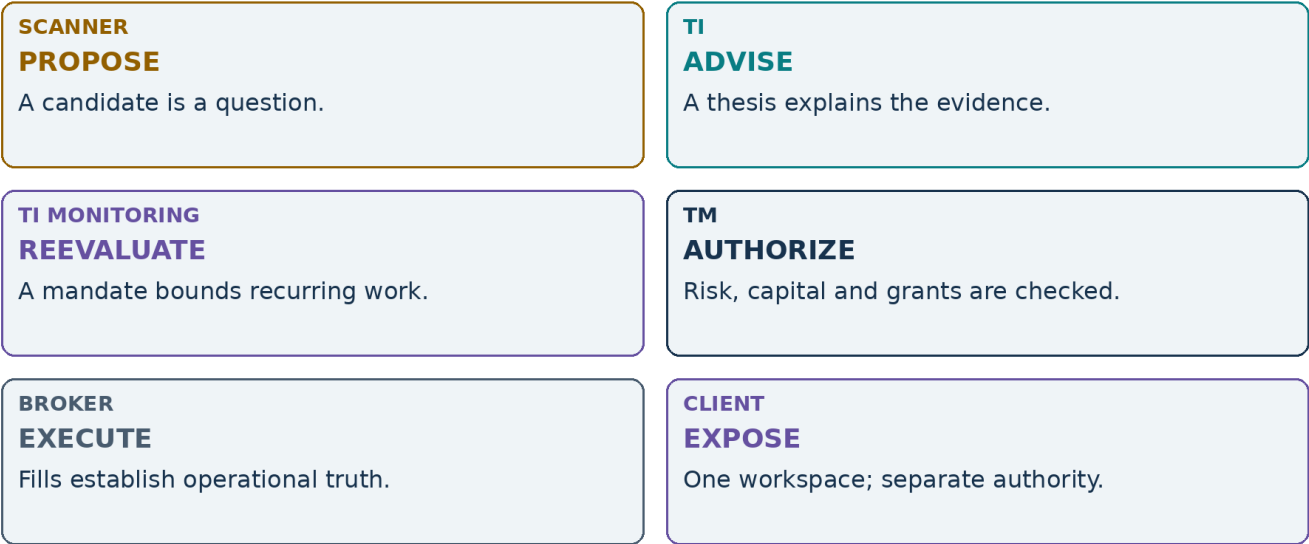


Figure 01. Six roles. One traceable workflow. No merged authority.

Scanners propose. TI advises. TI Monitoring reevaluates. TM authorizes and coordinates. Brokers execute. Human-facing clients expose and control the workflow without merging authority.

A human-readable companion to the authoritative Markdown architecture. All scenario trades, IDs, prices and charts are illustrative—not live validation, forecasts, recommendations or evidence of deployed integration.

READ WITH Architecture baseline: 12 September 2026 • Handbook: 13 September 2026 • Asia/Kolkata

ORIENTATION / READING GUIDE

What this handbook promises

This handbook teaches how an idea can become an accountable operational position without turning every component into a trading engine. Follow the objects, not just the arrows: each handoff changes who may decide, which evidence is required and what must be preserved for replay.

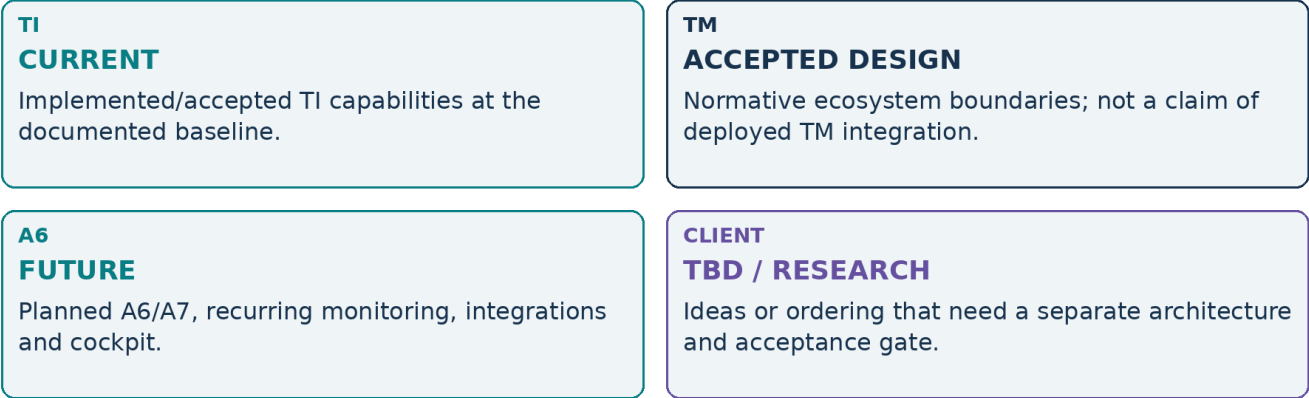


Figure 02. Read status words as well as owner colours.

Implementation truth

A1–A4 are frozen. A5.1/A5.2 and pluggability R1 are accepted; A5 freeze/tag readiness remains a separate gate. Public TI operations are local PURE/CAPTURED_READ capabilities. Replaying an input is not a fresh live evaluation. This thesis does not certify another TM repository or broker implementation.

A supportive thesis is not execution authority. Absence of required intelligence cannot silently become permission.

READ WITH TIAF_TRADING_ECOSYSTEM_ARCHITECTURE.md §1; TIAF_CAPABILITY_MAP.md

ORIENTATION / REFERENCE MAP

Contents & reading routes

Four chapters move from the mental model to safe handoffs, operator surfaces and concrete scenarios. Each teaching spread can also be read independently; owner labels and status labels travel with the visual.

Page	Chapter / reference	Focus
4	Chapter I	A map of responsibility
16	Chapter II	Handoffs that preserve authority
26	Chapter III	Surfaces, observability and resilience
36	Chapter IV	Follow the objects through a day
46	Reference	Boundaries to carry into implementation
47	Reference	Source map & compact glossary

Choose a route

Reader	Start here	Then inspect
New to the ecosystem	Chapter I: map, planes and state	Chapter IV: KAYNES and outage examples
Operator / product designer	Chapter III: tables and cockpit	Chapter II: authority and unknown outcomes
Implementer / reviewer	Chapter I: object ownership	Chapter II: admission; audit; pluggability
Architecture maintainer	Chapter IV: evolution and boundaries	Source map and normative Markdown

Solid arrows show information or governed requests—not automatic permission. Mockups are future concepts. “Illustrative” labels mean constructed teaching examples, never measured performance.

The handbook uses native Word headings, searchable text, editable tables and descriptive image alt text. Wide operational tables are split into linked column groups so their meaning remains readable on a page.

CHAPTER I / CHAPTER DIVIDER

A map of responsibility

Understand the ecosystem before following a trade.

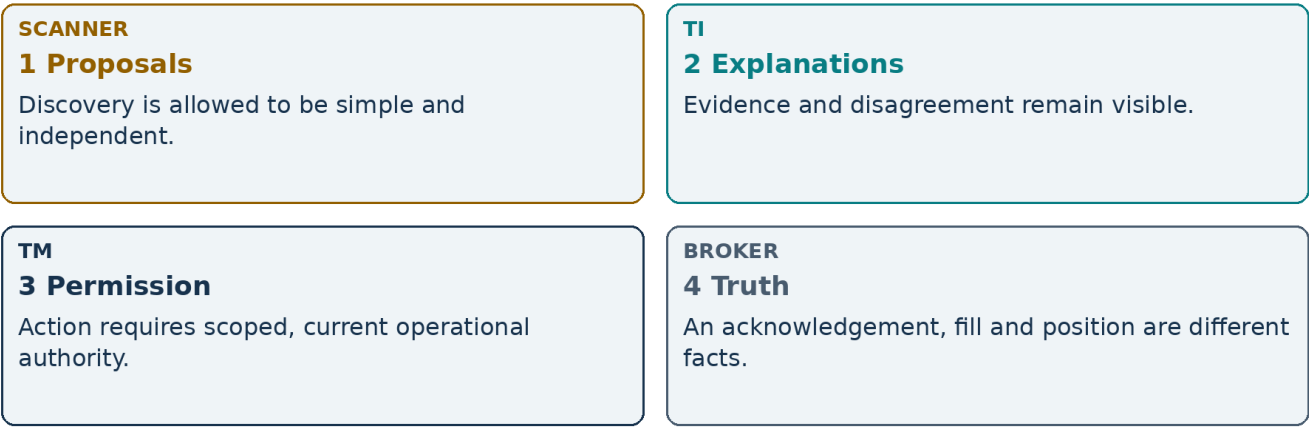


Figure 03. The four questions behind every handoff.

Do not ask one status light to explain discovery, intelligence and execution.

A system can correctly show an active bullish discovery, TI WAIT, TM BLOCKED and no broker exposure at the same time. The architecture preserves that combination rather than forcing a single flattering label.

I • THE ECOSYSTEM / ACCEPTED DESIGN

The full ecosystem

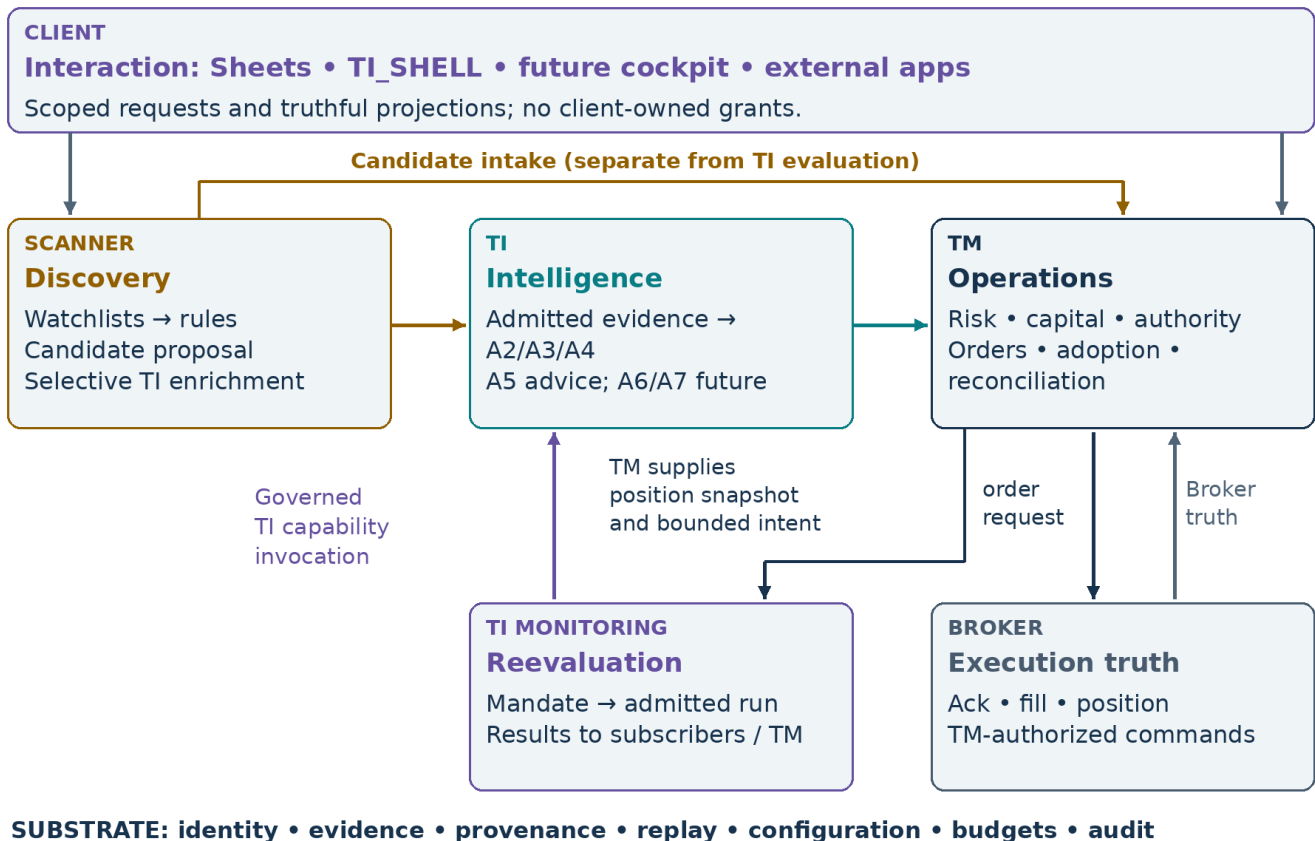


Figure 04. The ecosystem separates discovery, advice, operations and reevaluation.

The horizontal discovery-to-action path is only one route. TI may serve a research app without TM. A manual broker position enters through explicit TM adoption. A candidate can stop at WAIT or NO_TRADE. A monitoring mandate can reevaluate an opportunity even when there is no position.

The TI side may acquire authorized factual market data through adapters. That does not grant submit/modify/cancel authority. The broker box represents external execution and observed order/fill/position truth; TM reconciles it rather than creating truth from a cached display.

READ WITH TIAF_TRADING_ECOSYSTEM_ARCHITECTURE.md §§2,5,7-8

Five planes, not five services

LOGICAL PLANES • NOT FIVE REQUIRED SERVICES

Client	05 Interaction Console, Sheets and future UI invoke the same governed contracts.
Scanner	01 Discovery Universe selection, discovery refresh, candidate proposals.
TI	02 Intelligence Evidence admission, baseline, specialists, arbitration and advice.
TM	03 Governance & execution Risk, grants, capital, reconciliation; broker dispatch only here.
TI Monitoring	04 Intelligence orchestration Inside TI: admit mandates, schedule bounded runs, report freshness.

Figure 05. Logical separation does not prescribe a deployment topology.

Cross-cutting substrate	What it preserves
Identity + authorization	Caller, source, tenant/account/environment, rights and grants
Evidence + provenance + replay	Decision-time facts, source meaning, lineage and captured inputs
Configuration + pluggability	Exact policy/capability bindings; explicit required absence
Budgets + audit	Bounded work, attempts, unknown usage, traceable decisions

One local process may implement several logical planes behind explicit contracts. A future remote deployment adds authentication, persistence and operational guarantees; it does not dissolve the same boundaries. TM operational monitoring belongs in governance/execution, not inside TI Monitoring.

READ WITH *Ecosystem §2; TIAF_DEPLOYMENT_ARCHITECTURE.md*

I • THE ECOSYSTEM / ACCEPTED DESIGN

Three kinds of monitoring

SCANNER**Discovery refresh**

Which instruments satisfy this version of the discovery rule now?

TI MONITORING**Intelligence reevaluation**

Has admitted evidence changed this mandate's assessment?

TM**Operational monitoring**

What exposure exists, and what action is authorized now?

Figure 06. Similar clocks do not imply the same responsibility.

A scanner refresh can create, revise or expire candidates. A TI run can revise an explanation without creating any order. TM must still reconcile fills and exposure if the scanner or TI is unavailable. Operational continuity is bounded by available broker truth and existing authority—not a guarantee that every protective action will succeed.

Watchlist is organization; mandate is semantics. Monitoring is intelligence orchestration, not execution.

Current A5 WatchMandate/MonitoringNeed outputs are advisory intent. They do not create a scheduled job. The recurring TI Monitoring Runtime and its admission/budget/lifecycle guarantees remain future implementation work.

READ WITH *Ecosystem §8; TIAF_MONITORING_ARCHITECTURE.md §§1–8*

I • THE ECOSYSTEM / CURRENT + FUTURE

The TI intelligence hierarchy

TI	A1 Evidence foundation [CURRENT] Provider-neutral factual admission, provenance and captured inputs.
TI	A2 Deterministic baseline [CURRENT] Explainable benchmark; NO_TRADE and component evidence preserved.
TI	A3 Specialist intelligence [CURRENT] Specialists, bounded gateways, structured opportunity intelligence.
TI	A4 Challenge / arbitration [CURRENT] Preserve conflict, required absence and non-authorizing disposition.
TI	A5 Single-position intelligence [CURRENT] Advice over supplied position truth; not an execution or scheduling engine.
A6	A6 expression / A7 forecasting [FUTURE] Supported option expression; separately calibrated, non-authorizing overlay.

Figure 07. Capability roles, not a compulsory run of every layer for every request.

A3/A4 enrichment does not rewrite the accepted A2 record. The baseline remains an independently replayable comparison point. A5 interprets a single supplied open equity, future, call or put position; multi-leg shapes remain unsupported. Missing thesis or stale operational truth must not become a reassuring default recommendation.

READ WITH TIAF_CAPABILITY_MAP.md; A4/A5 architecture; TI_Intelligence_Hierarchy_Thesis.docx

I • DISCOVERY / ACCEPTED DESIGN

Basic scanner vs TI-enriched scanner

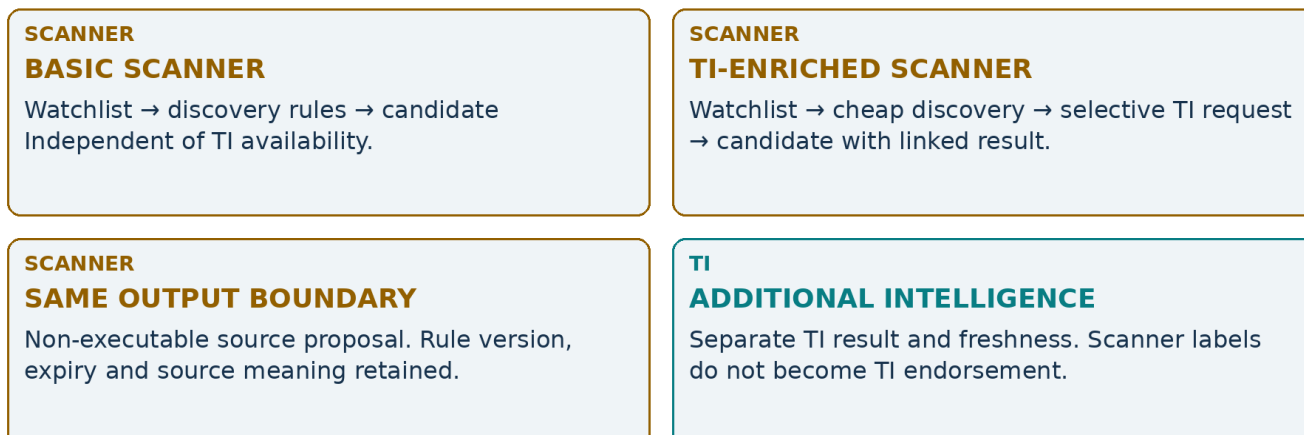


Figure 08. Enrichment adds a linked explanation, not a new source of authority.

TradingView, Chartink, Python, manual and future ML scanners can sit behind one candidate-source interface. A TI-native scanner is also a consumer application; it does not acquire extra Core privileges. Cheap screening followed by selective bounded enrichment avoids requiring deep research across an entire universe.

The subscriber owns watchlist membership and optional group labels. A candidate records the relevant group revision rather than consulting today's watchlist during replay. Discovery refresh and candidate expiry remain source responsibilities; neither implies an admitted recurring TI mandate.

READ WITH *Ecosystem §3*

Candidate: a versioned question

ILLUSTRATIVE FIELD VALUES • FUTURE CONTRACT DESIGN

SCANNER

IDENTITY

source=scanner-A • candidate=C17 • revision=2
predecessor=C17-r1 • rule=breakout/v3

SCANNER

SUBJECT & INTENT

KAYNES • canonical venue/type identity
DAY • opportunity • group=app1/watch3-r7

SCANNER

SOURCE CONTENT

BULLISH • score=72/100, rule points
Evidence refs + optional nested expression

TI

ADMISSION BOUNDARY

Origin authentication ≠ fact validation
Normalization ≠ admitted TI evidence

Figure 09. Readable field groups; not a published wire schema or a live KAYNES signal.

Required record groups	Meaning
Schema / event / correlation	Supported schema version, stable originating event, mapped trace context
signal_at / received_at / valid_until	Source time, receiver time and finite validity; late receipt does not reset expiry
Provenance + evidence references	Empty collections require explicit absence; raw source assertions remain attributable
Optional score / confidence / expression	Declared basis and scale; unknown semantics stay unknown; zero is valid

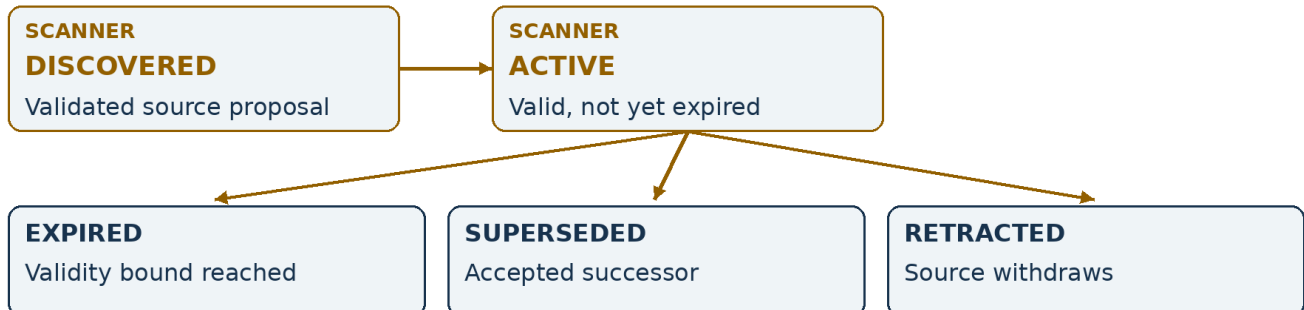
All timestamps are aware, normalized with zoneinfo.ZoneInfo("Asia/Kolkata") and serialized with +05:30. Naive inputs are rejected. Semantic collections remain immutable tuples with JSON arrays; extensible metadata is not promised deep immutability.

READ WITH Ecosystem §3; package version 0.1.0 is separate from A0 contract schema version 1.0

I • STATE / ACCEPTED DESIGN

Discovery and intelligence stay separate

DISCOVERY LIFECYCLE • SOURCE-OWNED



TI RUN STATUS ≠ TI DISPOSITION



ACTIVE + TI WAIT + TM BLOCKED is a coherent combined view.

Figure 10. Run completion is not a market opinion; an opinion is not an order.

A failed evaluation records a failed run. It cannot invent WAIT, SUPPORTIVE or NO_TRADE as if evidence supported that disposition. Conversely, a successfully completed A4 run may legitimately conclude ABSTAIN or INSUFFICIENT_EVIDENCE. UI status and domain meaning must both remain visible.

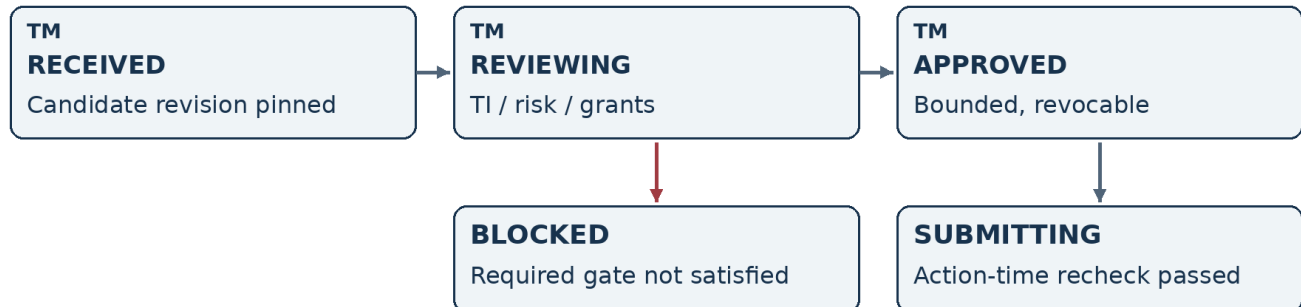
A source correction creates a successor revision. Retraction and expiry are retained events, not deletion. A superseded candidate invalidates linkage to old approval inputs, but does not silently cancel an already submitted broker order. TM reconciles that independent operational lifecycle.

READ WITH Ecosystem §§3–4

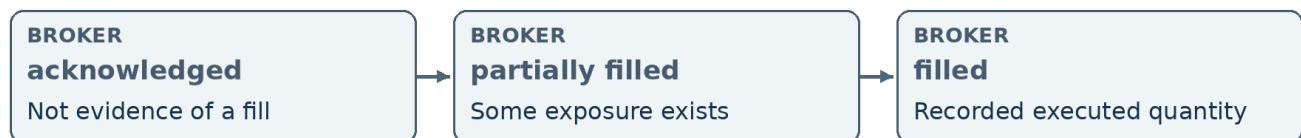
I • STATE / ACCEPTED DESIGN

Approval, submission and fill are different

TM PROPOSAL REVIEW



BROKER OUTCOMES • SEPARATE EVENT STREAM



rejected / cancelled / OUTCOME_UNKNOWN remain explicit.

Cancellation does not erase prior fills. Unknown submission outcome requires reconciliation before any retry; approval is not a fill.

Figure 11. The broker stream is not a continuation of a TI opinion enum.

An APPROVED UI row means a bounded TM decision existed—not that the action remains valid now. New exposure, revocation, expired evidence or a revised expression can invalidate dispatch. An acknowledged order is not yet a fill; a partial fill creates real exposure even if the remainder is cancelled.

Broker truth outranks cached assumptions. OUTCOME_UNKNOWN is not permission to submit a second order.

READ WITH Ecosystem §§4,7

The semantic object chain

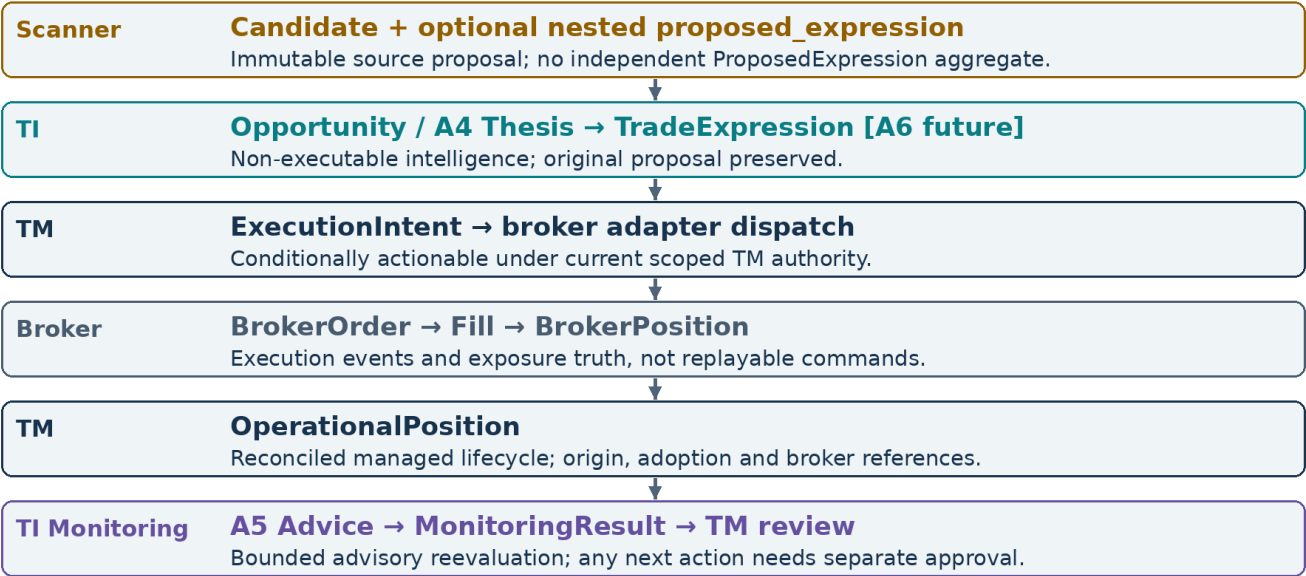


Figure 12. Different objects answer different questions; no mandatory TradeIntent is introduced.

This is a semantic map, not a compulsory pipeline. A NO_TRADE candidate stops. Equity need not pass through an options-only A6. Manual exposure enters through adoption. A broker position may aggregate several fills/orders; TM must preserve that mapping instead of assuming one candidate equals one position.

READ WITH Ecosystem §§4–5

I • OWNERSHIP REFERENCE / ACCEPTED DESIGN

Who owns discovery and advice?

The compact matrix separates the object owner from its producer and consumer. “No” in the authority column means the object cannot itself authorize execution. Immutable results may link to successors; a new interpretation never rewrites the earlier one.

Object / owner	Producer → consumer	Executable / authority	Persistence & replay
Watchlist Subscriber app	User/app → scanner / TI client	No; organization only	Version membership; retain admitted group context, no live membership lookup.
Candidate Source app	Authenticated source adapter → TI / TM	No; source proposal	Immutable revision + receipts/events; replay without rescanning.
Nested proposed expression Source app	Source within Candidate → A6 / TM	No; untrusted proposal	Version with Candidate; preserve original content and digest.
Opportunity / Thesis TI	A2/A3/A3.9/A4 → apps / A6 / TM / A5	No; evidence-based advice	Preserve child schemas, policy and captures; successors link to parents.
TradeExpression TI / future A6	Supported evaluator → TM	No; suitability ≠ permission	Pin candidate/input/alternatives; new result on reassessment.
A5Advice TI	A5 → subscriber / TM	No; protection/exit intent is advisory	Accepted capture/replay + policy identity; do not rename the runtime schema.

An object can be authoritative about its own meaning without being authorized to cause a broker action.

For example, TI owns the exact A4 conclusion and its evidence lineage. TM cannot relabel WAIT as SUPPORTIVE. But only TM can decide whether a separately specified policy permits an operational action, using current grants, account state and risk constraints.

READ WITH *Ecosystem §5*

I • OWNERSHIP REFERENCE / ACCEPTED DESIGN

Who owns action and reevaluation?

Object / owner	Producer → consumer	Executable / authority	Persistence & replay
ExecutionIntent TM	Authorized TM workflow → TM broker adapter	Conditional; current TM gate required	Durable action ID, pinned approval and attempts; amendments reauthorized; replay never dispatches.
BrokerOrder Broker	Broker → TM / projections	Actual order state; not client-authored truth	Broker IDs, updates, acknowledgements and unknowns retained; no replay submission.
Fill Broker	Broker → reconciliation	Executed fact, not reusable permission	Observed ID/account/qty/price/time; append attributable corrections.
BrokerPosition Broker	Snapshots/events → TM	Exposure truth, not command	Version observations and acquisition times; preserve uncertainty.
OperationalPosition TM	Reconcile/adopt → risk / TI / clients	Managed lifecycle under scoped grants	Stable account/position ID, origin and revisions; no live broker call in replay.
MonitoringMandate TI after admission	Subscriber intent → TI runtime	Bounded intelligence only	Admitted revisions + lifecycle; retain original A5 intent if supplied.
MonitoringResult TI	Capability run → subscribers / TM	Advisory, never permission	Mandate/run/attempt/input/ policy/usage links; offline replay, no reacquisition.

Each owner persists its records and cross-links; a shared database is not required. Retention must respect rights and safety. Active actions, unknown outcomes and unresolved usage cannot disappear before reconciliation. Exact durations, storage engine and export protocols remain implementation gates.

READ WITH *Ecosystem §5; Monitoring §§10–11*

CHAPTER II / CHAPTER DIVIDER

Handoffs that preserve authority

An end-to-end sequence is a chain of checks—not a green-light cascade.

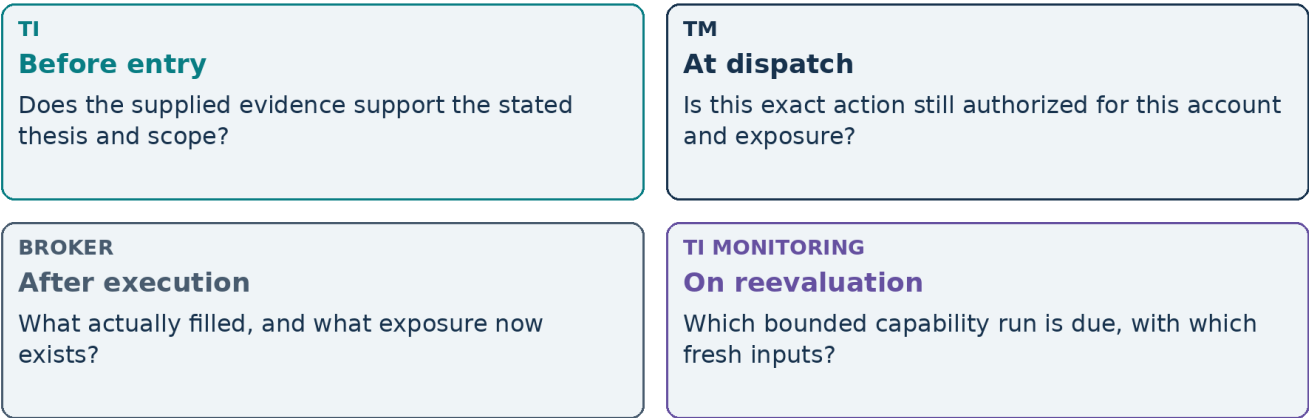


Figure 13. The question changes at every handoff.

TI advises on meaning. TM decides under authority. A6 expression quality cannot replace TM’s capital, quantity or account checks.

Future integration must demonstrate the complete trace, including blocked, partial and unknown outcomes. Successful-looking examples are insufficient if the same path becomes permissive under timeout or stale evidence.

II • BEFORE ENTRY / ACCEPTED DESIGN

Scanner → TI → A6 → TM → Broker

ACCEPTED DESIGN • NOT A LIVE INTEGRATION TRACE

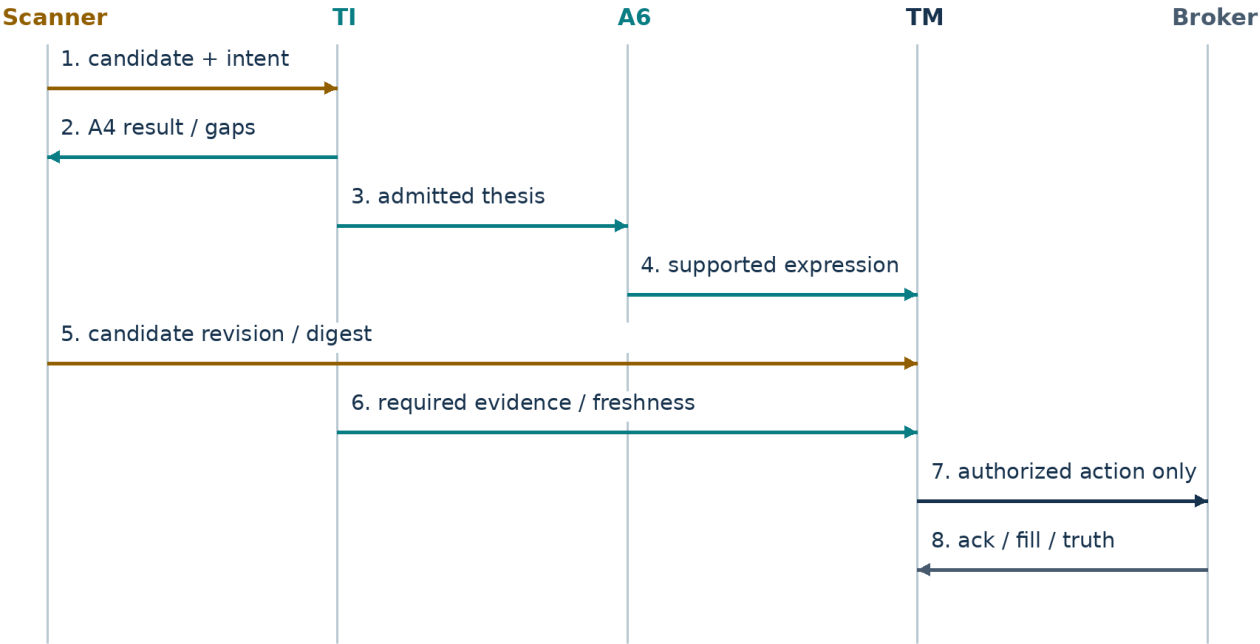


Figure 14. A6 is future; TI and source artifacts remain separately pinned.

TM may intake a candidate before evaluation finishes. What matters is the approval boundary: required intelligence must satisfy the exact declared policy before entry. The sequence does not imply every source invokes A6 directly, nor does it publish a remote API. Today’s captured-read facade cannot serve as proof of fresh live validation.

SUPPORTIVE → expression consideration. Not SUPPORTIVE → order.

READ WITH Ecosystem §§6–7; current facade boundary in §1

II • BEFORE ENTRY / FUTURE

The A6 boundary

SCANNER**EXTERNAL PROPOSAL**

Nested expression stays attributable and untrusted; source revision changes on edit.

A6**EVALUATED EXPRESSION**

Supported deterministic single-leg CE/PE only; alternatives, gaps and rejection reasons retained.

TM**OPERATIONAL DECISION**

Account eligibility, actual quantity, capital, grants and exact order parameters.

TI**A7 IS A SEPARATE OVERLAY**

Calibrated forecast/evaluation evidence may enrich later work; not needed for initial A6.

Figure 15. A6 answers “which supported expression?”, not “may this account trade?”.

A6 may compare admitted expiry/strike/moneyness, liquidity, IV/Greeks and accepted constraints. It may return NO_OPTION_TRADE. Exact eligible instruments, exposure side permissions, orderability rules and thresholds require separate acceptance. No spread optimizer, hedge/roll engine or naked-short permission is implied.

A cash or future proposal can use an independently approved TI/TM path; it must not pretend options-only A6 endorsed it. No probability-of-profit or expected-move forecast is fabricated to fill missing evidence.

READ WITH Ecosystem §6; authoritative roadmap A6/A7

II • BEFORE ENTRY / ACCEPTED DESIGN

Required scope cannot disappear

TM
TI_REQUIRED
Present + complete + fresh + acceptable for the declared required scope. Otherwise block new entry.

TM
TI_OPTIONAL
Only a preapproved versioned strategy may proceed without validation. Record NOT_TI_VALIDATED.

Figure 16. Policy selects the mode before the request; an outage cannot select a weaker mode.

Approval binds	Action-time recheck
Candidate revision / semantic digest; selected expression	Same proposal and approved parameter envelope; changed input requires revalidation
TI result/A4 IDs, hashes, evidence cutoff and expiry	Freshness and required completeness still acceptable; recorded veto not hidden
Actor / account / environment / grants / policy versions	Grant still valid; risk, capital and broker exposure/open orders current
Captured composition where available; parent links	COMPOSITION_UNAVAILABLE stays explicit; a strict composition requirement blocks

Missing optional enrichment need not block a complete required scope. Missing required capability remains part of completeness under R1. An unfavorable answer is not an outage; TI_OPTIONAL cannot relabel a known applicable veto as missing data.

READ WITH Ecosystem §7; TIAF_PLUGGABILITY_ARCHITECTURE.md; R1 accepted behavior

II • AFTER EXECUTION / ACCEPTED DESIGN + CURRENT A5

The post-fill A5 monitoring loop

ACCEPTED DESIGN • NOT A LIVE INTEGRATION TRACE

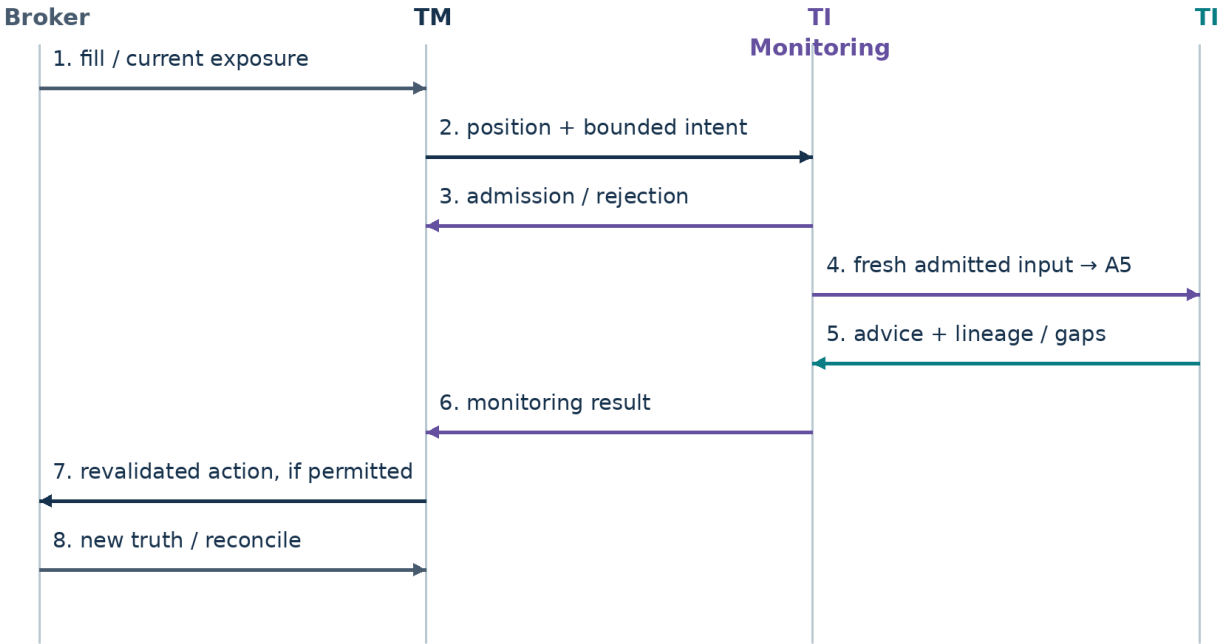


Figure 17. Recurring dispatch is future; A5 advice is already a bounded, non-executable capability.

A5 can emit MAINTAIN, WATCH_CLOSELY, PROTECT, REDUCE_RISK, TRAIL_PROTECTION, EXIT_RECOMMENDED, WAIT_FOR_CONFIRMATION, ABSTAIN or INSUFFICIENT_EVIDENCE. These are recommendations, not order types. A stop/protection intention does not prove a protective broker order exists.

TM tracks what actually executed, including partial reductions and failures. A later A5 run consumes a new supplied operational snapshot; it does not mutate TM state or assume the earlier recommendation was implemented.

READ WITH Ecosystem §8; A5 architecture §§4–7; Monitoring §§5–8

One mandate, five distinct clocks

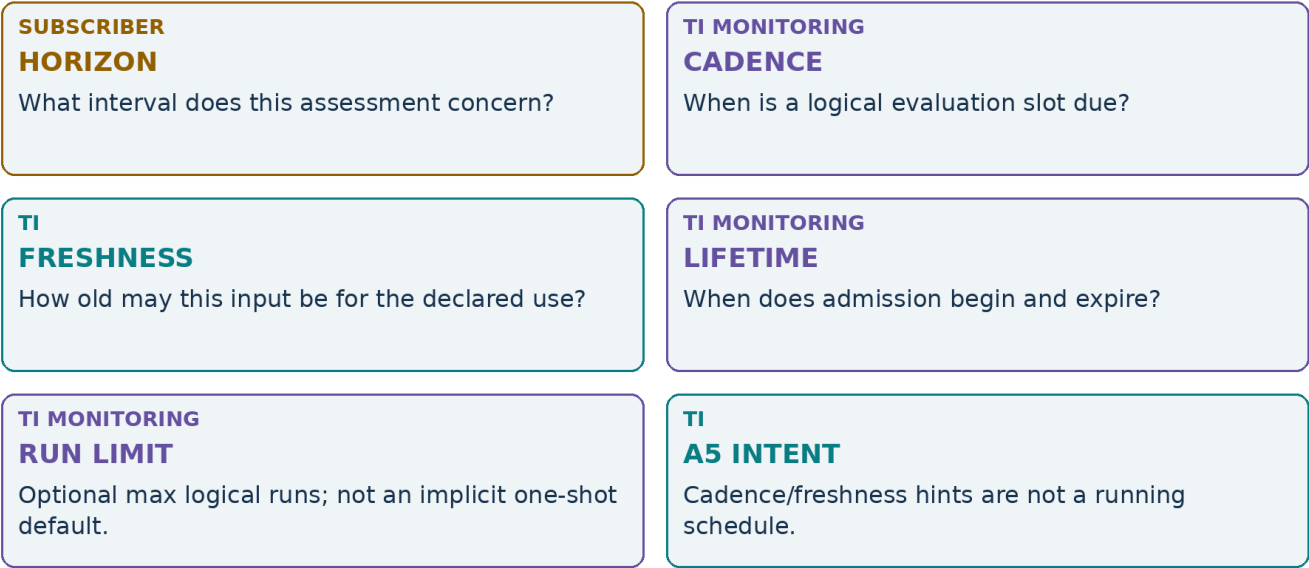


Figure 18. Horizon is not cadence; a finite lifetime does not specify a run count.

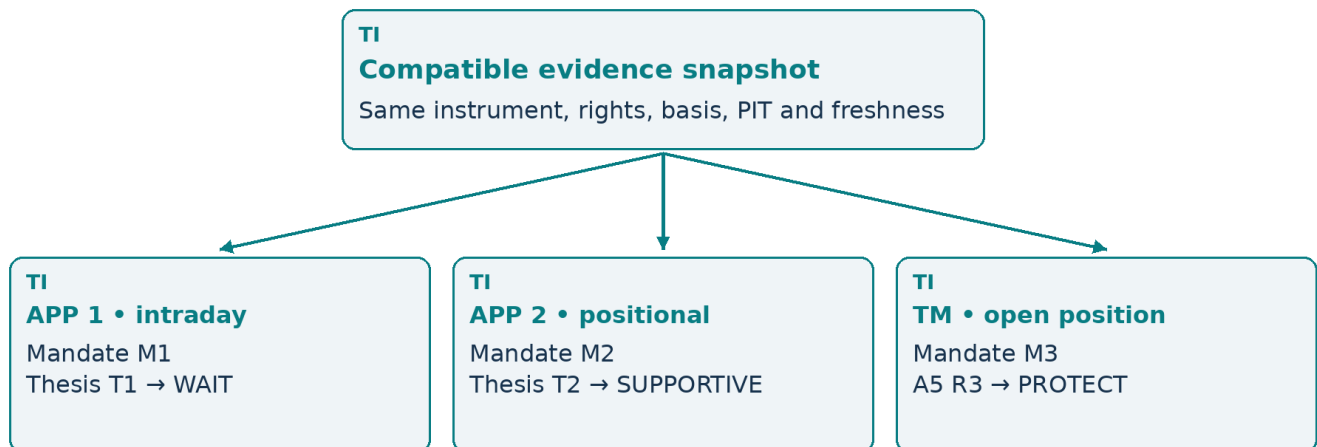
A subscriber submits intent; TI admits or rejects a bounded mandate against identity, rights, profile, capabilities, budgets and capacity. A watchlist is optional organization. One instrument may carry multiple mandates with different purposes, horizons and rights.

The planned local runtime uses bounded due slots and a single writer, with at-least-once delivery and idempotent logical run IDs. Retries retain separate attempts. Missed slots may be coalesced under accepted policy; no backdated “fresh” result or silent lifetime renewal is allowed. Unknown attempt cost remains unknown and conservatively reserved.

READ WITH `TIAF_MONITORING_ARCHITECTURE.md` §§3-8,10-11

II • REUSE / ACCEPTED DESIGN

Shared evidence, separate analysis



Shared evidence ≠ shared analysis state

Intent, purpose, horizon, position, policy and conclusions stay separate.
A cache hit is never permission to reuse another subscriber's thesis.

Figure 19. The same KAYNES evidence can support different scoped conclusions.

A safe evidence reuse key includes instrument/venue, provider and account entitlements, requested fields, basis, adapter/schema/policy versions, point-in-time cutoff, availability and each consumer's freshness. Unknown dimensions are not wildcards. Access rights must still hold when returning a cached item.

Analysis state also depends on purpose, horizon, position, policy and subscriber intent. It is not safe to copy APP 2's supportive thesis into APP 1's intraday request merely because the instrument matches. Nor should shared source material be counted twice as independent corroboration.

Deduplicate evidence, not intent. Reuse is an optimization of compatible work—not a merge of meaning.

READ WITH Monitoring §9; source-authority architecture; ecosystem §§3,8

II • WORKLOAD / ILLUSTRATIVE CHART

A day in the life of shared evidence

ILLUSTRATIVE • NOT MEASURED PRODUCTION DATA

7 apps → 14 watchlists → 183 active intents → 96 instruments



71 repeated acquisitions avoided in this constructed round (38.8%).

Assume one evidence family per intent; entitlement/freshness splits create 112 groups, not 96. Real savings depend on compatibility.

Figure 20. A constructed one-round workload; acquisition savings do not reduce intent count.

Assume all 183 active intents require one comparable evidence family in this round. Ninety-six instruments do not guarantee ninety-six safe acquisitions: entitlement and freshness boundaries split some instruments into multiple groups. For teaching, suppose this produces 112 compatible groups. Then $183 - 112 = 71$ repeated acquisitions can be avoided.

This arithmetic does not measure deployment throughput, latency, cost or provider savings. A real workload contains multiple evidence families, cache misses, changing rights and refresh windows. The 183 purpose-specific analyses remain separate even when they share an admitted input snapshot.

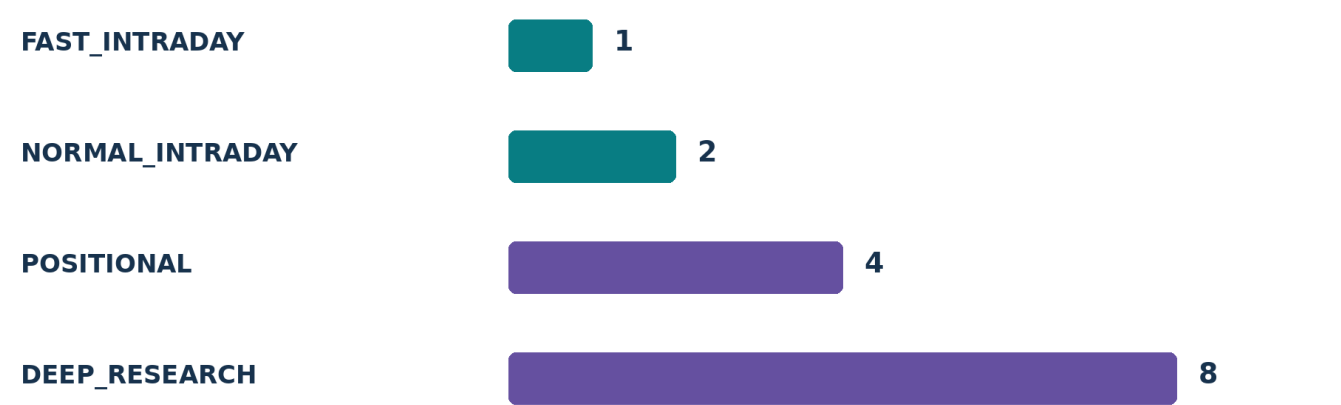
Fewer fetches does not mean fewer responsibilities.

READ WITH Monitoring §§9,11; ecosystem profiles and budgets

II • BOUNDED WORK / ILLUSTRATIVE CHART + ACCEPTED DESIGN

Latency and cost are policy dimensions

ILLUSTRATIVE BUDGET INDEX • NO TIME / COST SLA



Index values are teaching aids, not implemented settings or benchmarks.

Figure 21. Conceptual resource envelopes, not runtime defaults or measured performance.

Each versioned profile declares	Why it matters
Required / optional intelligence	A deadline can drop optional work only under policy; required scope is never silently shrunk.
Freshness + deadline	Input age and completion time are different constraints; stale success is not fresh success.
Budget + failure behavior	Bound acquisition/model/attempt costs; return explicit partial/blocked outcomes.

FAST_INTRADAY, NORMAL_INTRADAY, POSITIONAL and DEEP_RESEARCH name conceptual usage profiles. No numeric SLA is promised here. Real latency would include acquisition, queueing, execution and retries; real cost needs observable usage. Unknown usage is not zero, and a budget exhaustion cannot be made invisible by excluding an attempted call.

READ WITH *Ecosystem §12; Monitoring §11*

II • ADOPTION / ACCEPTED DESIGN

A broker position is not yet a managed position

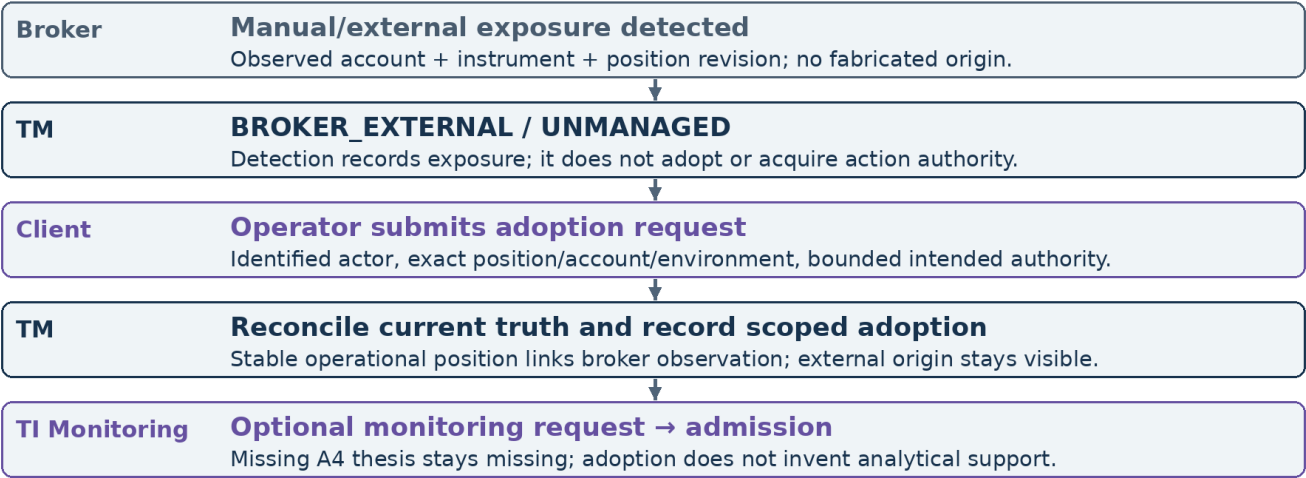


Figure 22. Detection ≠ adoption ≠ thesis creation ≠ execution permission.

A stale UI row is not enough to adopt the latest position revision. TM checks the exact account, current quantity and authority scope, then records who adopted what under which policy. An already managed position must not be adopted twice through duplicate client retries.

If a valid A4 thesis is absent, current A5 may return `INSUFFICIENT_EVIDENCE`. If current operational truth is stale or unknown, `ABSTAIN` is the safe intelligence outcome. TM still owns whatever operational controls its existing grants and available truth permit.

READ WITH *Ecosystem §8; A5 architecture §§3,11*

CHAPTER III / CHAPTER DIVIDER

Surfaces, observability and resilience

Make the workflow legible without relocating its authority into the screen.

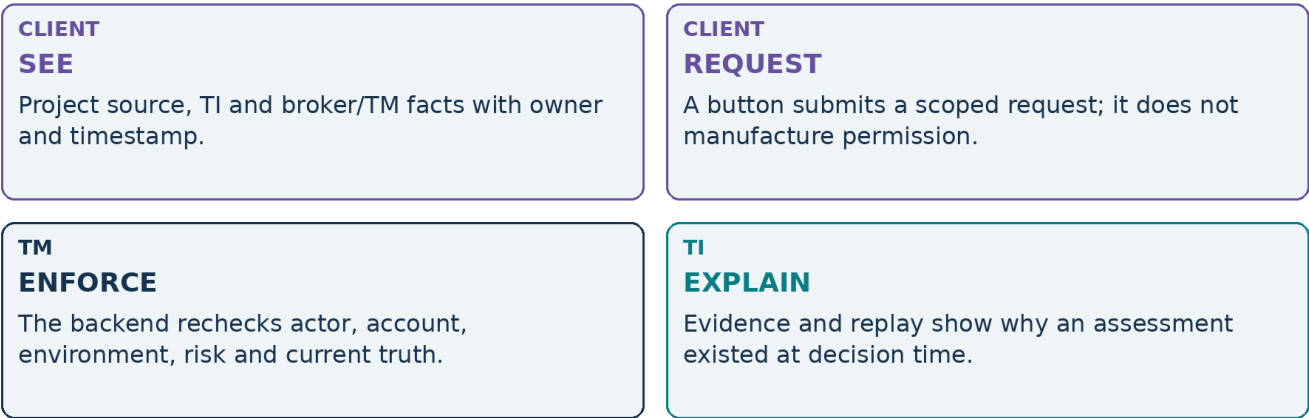


Figure 23. The interface is a lens and a request surface, not a parallel backend.

One cockpit must not mean one undifferentiated “approved” state.

The next pages deliberately split wide tables into linked bands. A compact daily-use grid needs concise defaults and expandable evidence, not dozens of unreadable columns or a single green traffic light that conceals conflicts.

III • CLIENTS / FUTURE MOCK LAYOUT

Google Sheets: replaceable client



Figure 24. A Sheet edit changes allowed subscriber data; an action cell sends a request.

Example tabs	Role / write boundary
Dashboard • Alerts	Read-only aggregates and freshness/health; acknowledge only through scoped request
Watchlists • Candidates	Subscriber membership / drafts editable; accepted revisions and TI/TM columns read-only
Opportunities • TI Advice	TI explanation, gaps, sources and timestamps; read-only
Active Positions • Orders • Trade Log	Reconciled broker/TM records; read-only
Config	Permitted preferences or change requests—not raw grants, risk bypasses or credentials

READ WITH Ecosystem §9; no broker keys/tokens in Sheets; no automatic replay of stale approvals

III • CANDIDATE MOCKUP / ILLUSTRATIVE FUTURE UI

One candidate row, several truths

These are two horizontally linked bands of the same conceptual grid. The row key pins source/candidate/revision. Detailed raw-score basis, sector context, qualification and forecast fields expand on demand; the compact view never blends them into one status.

Key / source / group	Instrument / horizon / source direction	Discovery / age / expiry	TI disposition / freshness
C17-r2 Python-A / Day-1	KAYNES / DAY BULLISH	ACTIVE 2m / 10:20+05:30	SUPPORTIVE as_of 10:02+05:30; fresh for demo scope
C22-r1 Chartink-B / Day-2	KAYNES / DAY BULLISH	ACTIVE 1m / 10:18+05:30	WAIT same-round admitted assessment
C31-r1 Manual / Swing	RELIANCE / POSITIONAL unspecified	ACTIVE 1d / explicit validity	NOT_EVALUATED no TI freshness claim

Same row key	Expression / owner	TM risk / authority / execution	Request actions
C17-r2	CE-demo-01 future A6 result	PASS / pending approval / not submitted	Inspect evidence; request review
C22-r1	Not evaluated	BLOCKED under TI_REQUIRED / no grant / none	Inspect WAIT reasons
C31-r1	Source equity proposal; not A6	Not reviewed / none / none	Request TI evaluation

The same symbol can appear twice because the sources and candidate identities differ. “TI fresh” is a scope-specific check, not a permanent badge.

Every time above is on the same hypothetical session date, 2026-09-14, and in Asia/Kolkata; dates must be retained in persisted records. These examples do not assert market prices, current instrument eligibility, or existing operational grants. Production controls must bind exact revision and current actor/account/environment.

READ WITH Ecosystem §§3–4,9

III • POSITION MOCKUP / ILLUSTRATIVE FUTURE UI

Operational truth beside TI advice

The row key joins these mock bands; the real UI can use expandable panels. All quantities, prices and P&L below are constructed examples for display semantics—not observed data. “Broker B / DEMO” is a fictional context. The position is a simple equity example, not an A6 options claim.

Position / owner	Exposure / source	Origin / adoption / as_of
Broker B / DEMO BP-EQ7 → P-EQ42-r3	RELIANCE equity Qty 10; entry ₹1,000; mark ₹1,010	BROKER_EXTERNAL; adopted by TM 2026-09-14T11:00:00+05:30
P&L / TM-derived display	₹100 unrealized = 10 × (1,010 – 1,000)	Fees excluded; mark timestamp retained Not a broker-confirmed realized return

TI-owned intelligence	TM-owned operations	TI-runtime health
A5 posture: CAPITAL_AT_RISK Thesis: INTACT Advice: MAINTAIN	Risk: within demo envelope Authority: scoped, expires Protective order: NOT_CONFIRMED	Runtime: NOT_ACTIVE Last captured A5: 11:00+05:30 Next due: unavailable
Protection intent: none newly supplied Evidence/gaps: expandable	Broker/account context pinned Actions: request review / reconcile	Cadence hint, if present: advisory Not proof of scheduled work

An analytical protection posture is not a protective broker order. “Last captured evaluation” is not a live monitoring heartbeat.

A5 recommendation, thesis health and analytical risk posture remain distinct fields. TM owns quantity, risk and action grants; broker observations anchor live exposure. If monitoring later exists, display last evaluation, next admitted due slot, expiry and overdue status independently of a stale cached recommendation.

READ WITH Ecosystem §9; A5 architecture §4; Monitoring §5

III • WORKSPACES / FUTURE UI • CURRENT CONTRACT BOUNDARIES

TI workspace vs TM workspace

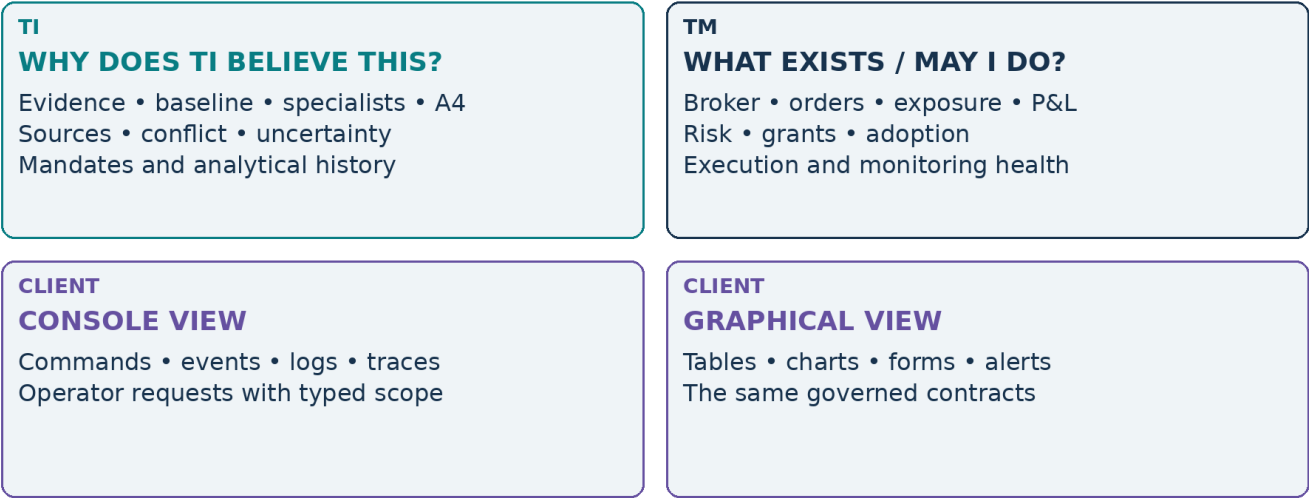


Figure 25. A console and a UI are two views of the same boundaries.

Current TI_SHELL is a local client of curated PURE/CAPTURED_READ operations. A future Web client or Sheets connector must not expand capability authority just because it has a convenient button. It also cannot infer successful adoption or order state from a submitted request.

A shared selection can synchronize KAYNES and a chosen candidate/position across both workspaces. Identity, permissions, provider rights and backend ownership still remain explicit. One panel’s successful request is not another panel’s permission grant.

READ WITH Ecosystem §9; deployment architecture; TBD CLI/Web UI ideas remain future

III • TRADING COCKPIT / FUTURE MOCK ARCHITECTURE

A multi-console daily workstation

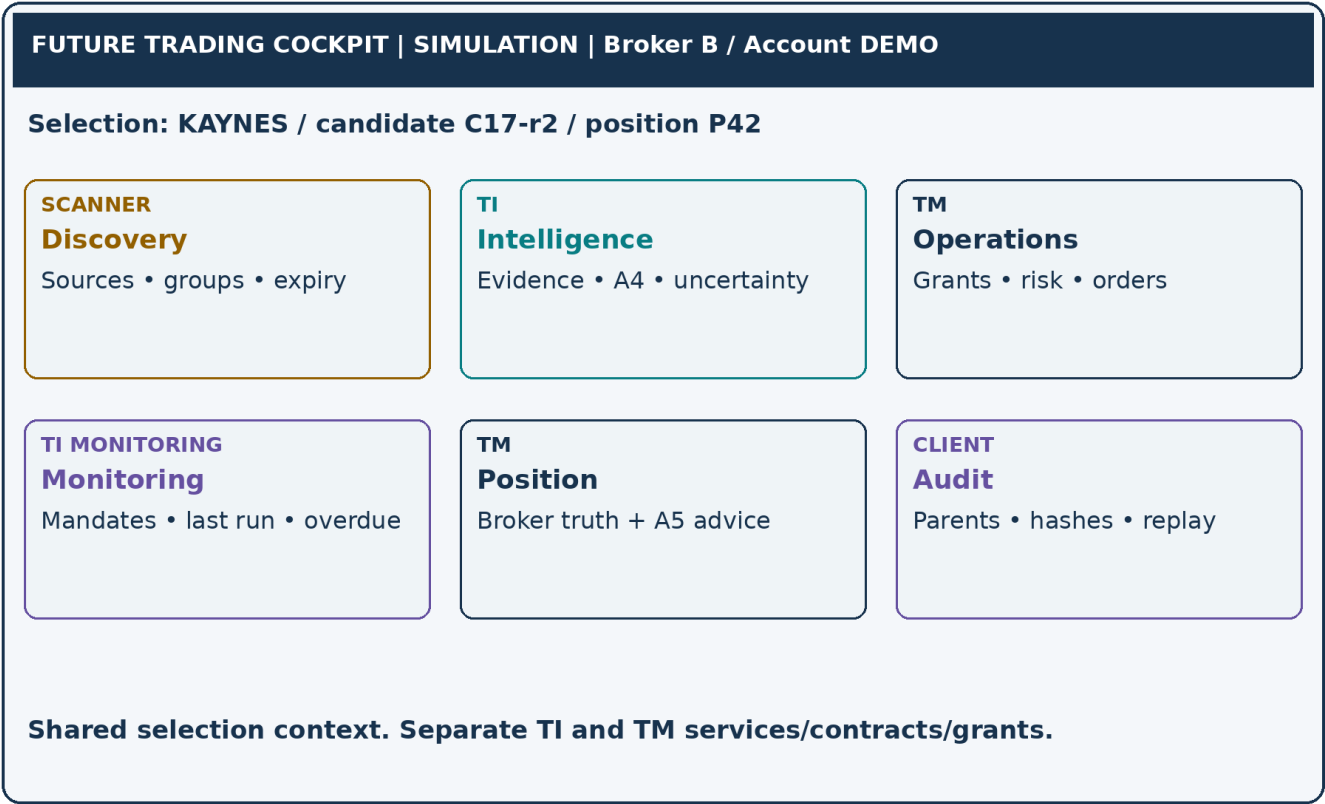


Figure 26. A cohesive workspace built from separate governed panels.

The selection context may carry canonical instrument, candidate revision, thesis/result ID and operational position ID. It is navigation state—not shared mutable analysis or a common authority token. Each panel requests only the scopes its backend and acting principal permit.

A role-limited research workspace may omit order controls entirely. An operations workspace must make broker/account/environment impossible to confuse and recheck authority server-side. UI outage must not freeze otherwise authorized backend reconciliation; stale reconnect requests must not auto-execute.

Unified experience. Separate ownership. Explicit account and environment.

READ WITH Ecosystem §9; future interaction technology and session design remain gated

III • AUDIT / ACCEPTED DESIGN

Why was this trade taken?

ILLUSTRATIVE IDS • EACH OWNER RETAINS ITS RECORDS

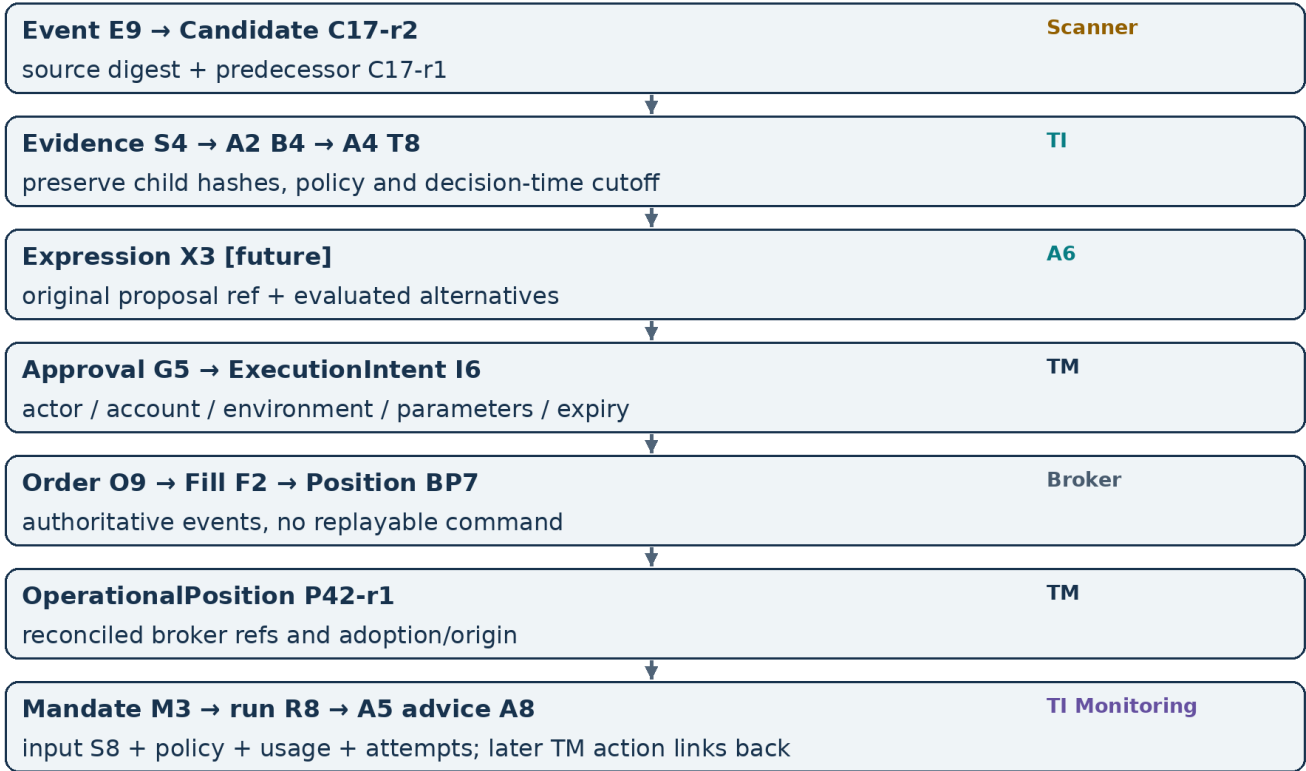


Figure 27. One correlation context links immutable local objects; it does not replace their IDs.

Keep event time, availability time, acquisition time and decision as_of distinct. Child evidence fingerprints, baseline and A4 outputs remain unchanged; later comparison runs are separate artifacts. One event may fan out to multiple candidates, subscribers, fills or mandates, so lineage is a graph, not one mutable “trade row”.

READ WITH Ecosystem §§5,10; A4/A5 replay; source-authority architecture

III • REPLAY / ACCEPTED DESIGN

Replay explains; replay never executes

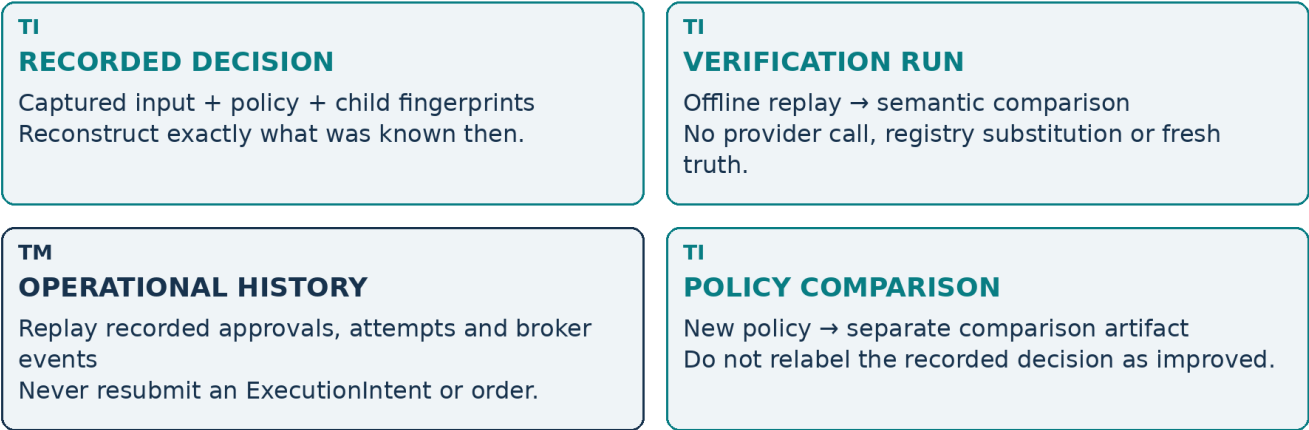


Figure 28. Captured reconstruction and new-policy comparison are different products.

A lineage bundle should retain the decision-time required scope, missing evidence, contradiction findings, source rights and usage observations alongside the successful outputs. A future full composition manifest cannot be reconstructed by consulting today’s registry; absent identity remains COMPOSITION_UNAVAILABLE.

Current captured replay is an implemented TI capability. Durable cross-system export, exact retention and TM replay integration still require their own acceptance. A saved research result is not automatically a current approval, and a successful replay says nothing about today’s broker exposure.

READ WITH Ecosystem §10; pluggability §§9–10; A3.10/A4/A5 captured replay

III • RESILIENCE / ACCEPTED DESIGN

Failure and degradation matrix

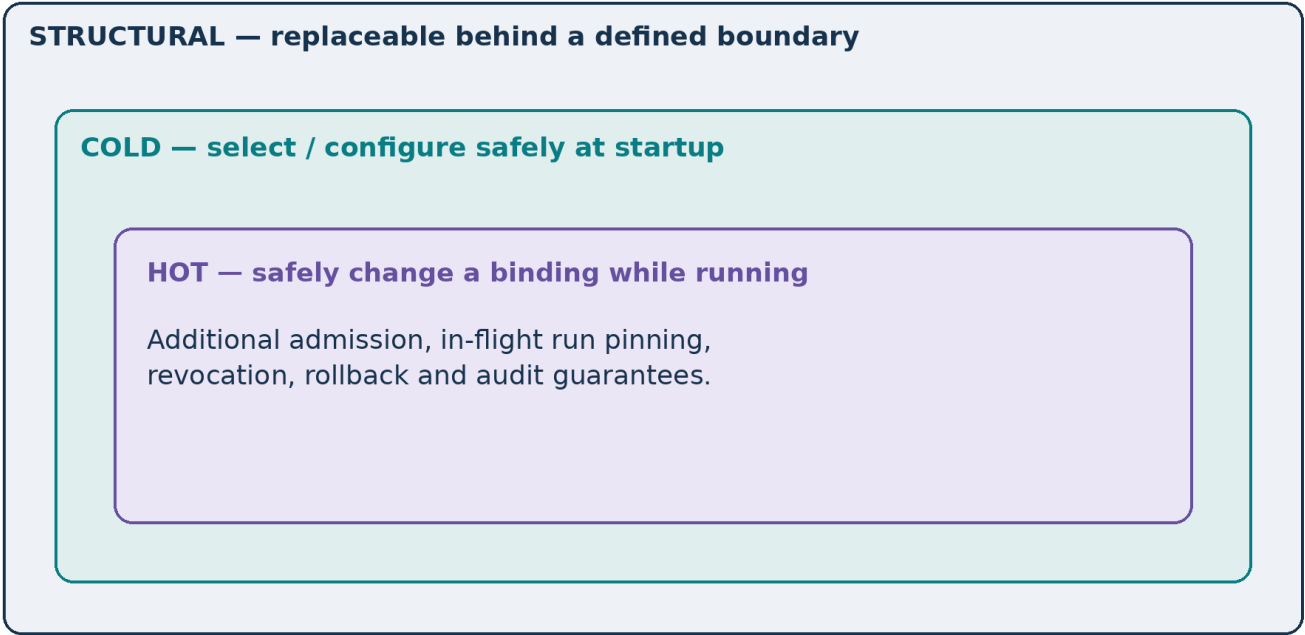
Failure	New-entry behavior	Existing work / position	Truth that must remain visible
Scanner unavailable	No fabricated new candidates; existing validity still expires	TM reconciles existing exposure; admitted TI work may continue within bounds	Source outage, candidate age and expiry
TI unavailable	TI_REQUIRED blocks; TI_OPTIONAL only under preapproved policy	TM's independent controls continue within existing authority and available truth	NOT_TI_VALIDATED if permitted; last TI as_of and gaps
Required capability absent	Block / explicit incomplete result; do not shrink required scope	Other safe, independent work can continue	Required absence and exact profile/version
Broker unavailable / timeout	Block uncertain new exposure; reconcile before retry	Do not promise protection can execute; preserve open/unknown states	OUTCOME_UNKNOWN; attempts and last broker truth
Sheets / UI unavailable	No unverified queued approvals on reconnect	Backend may continue admitted bounded work and reconciliation	Client stale/disconnected; backend status separately
TI Monitoring unavailable	No claim of fresh recurring validation	TM operational controls remain independent; advice can become overdue	Last actual evaluation, expiry, overdue / NOT_ACTIVE
Budget exhausted / unknown usage	Explicit partial/blocked result per policy; required work not silently removed	Keep reservations and attempt lineage for reconciliation	Observed vs unknown usage; remaining admissible budget

Failure isolation is not fail-open permission. “Continue managing” means within current rights and observable truth—not guaranteed execution.

READ WITH *Ecosystem §11; Monitoring §§8,11; pluggability §11*

III • PLUGGABILITY / ACCEPTED DESIGN

Replace components, preserve meaning



HOT ⇒ COLD ⇒ STRUCTURAL; not the reverse. Optionality is a separate axis.

Figure 29. Stronger replacement guarantees contain weaker ones; requiredness is separate.

Component family	Boundary to preserve
Scanner / broker adapters	Source normalization vs TM execution adapter authority
TI capability / model provider	Declared scope, evidence semantics, bounded usage and replay identity
Scheduler / Sheets connector / UI panels	Admitted mandate semantics, scoped requests and faithful projections

No HOT requirement is imposed merely because a component is pluggable. R1 preserves required-scope absence today; broader R2–R5 work is still pending. A configured-but-unready component and an unavailable required capability are not equivalent to an optional component that was never requested.

READ WITH `TIAF_PLUGGABILITY_ARCHITECTURE.md` §§2–6,13–17

CHAPTER IV / CHAPTER DIVIDER

Follow the objects through a day

Six constructed scenarios test the mental model—including stops and failures.

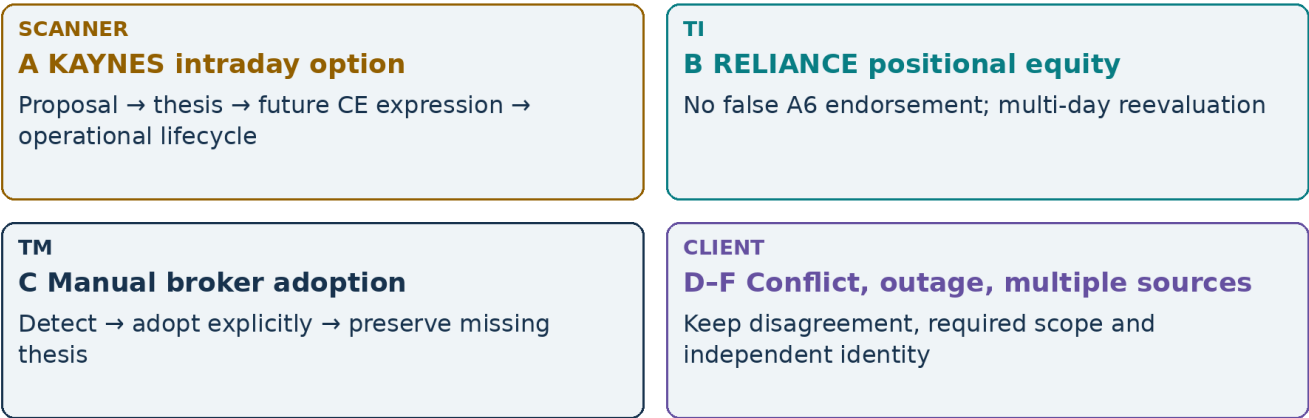


Figure 30. Worked cases explain the architecture; they are not market recommendations.

All case IDs, session times, market facts and quantities are illustrative. A named stock does not assert current eligibility, liquidity, listing availability or a live assessment.

The cases assume future integrations have passed their own acceptance gates. Current A5 advice and captured replay boundaries remain unchanged. Dates use a hypothetical session and Asia/Kolkata; no P&L trajectory or probability is forecast.

IV • CASE A / 1 / ILLUSTRATIVE FUTURE WORKFLOW

KAYNES: from proposal to filled exposure

HYPOTHETICAL SESSION 2026-09-14 • ALL TIMES +05:30

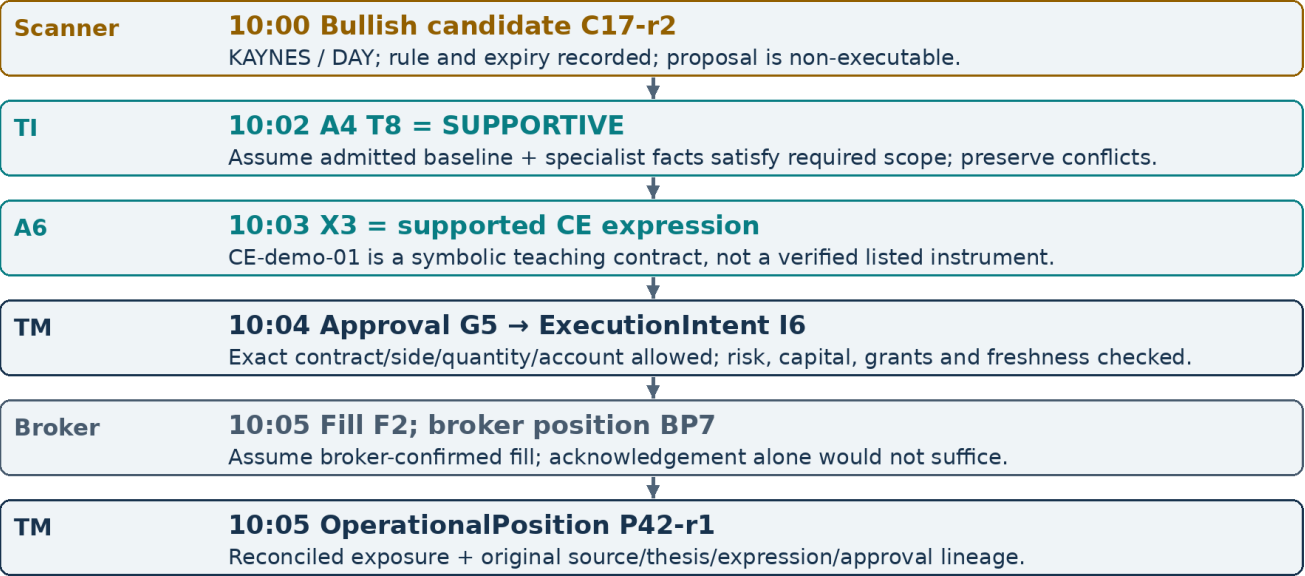


Figure 31. Every step names both the owning component and the object it produces.

Assume a future accepted A6 policy supports this exact instrument and exposure side; the thesis does not choose a strike, expiry, lot size or current long/short permission. TM remains responsible for actual quantity and account eligibility. Any failed required check stops entry even when the underlying thesis is SUPPORTIVE.

READ WITH Case continues on next page; ecosystem §§6–8

IV • CASE A / 2 / ILLUSTRATIVE FUTURE WORKFLOW

KAYNES: MAINTAIN → PROTECT → exit advice

CONSTRUCTED ADVICE SEQUENCE • NOT A REQUIRED STATE MACHINE

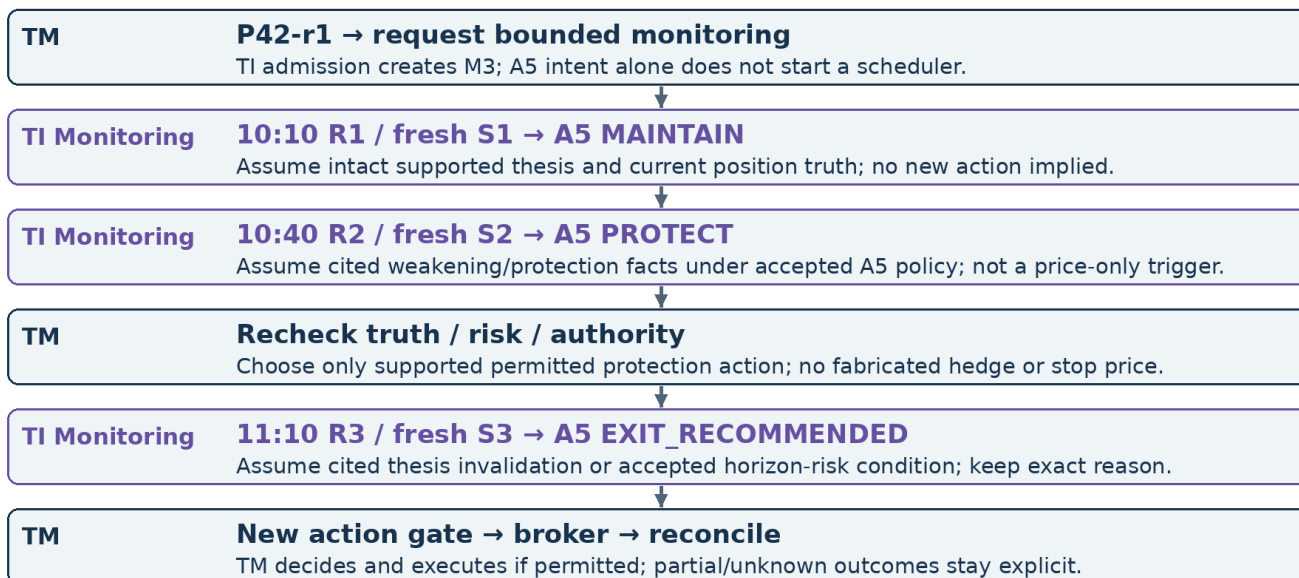


Figure 32. Each advice result is a new A5 run over new supplied truth, not an automatic transition.

At any evaluation, stale/unknown truth can produce ABSTAIN; missing thesis can produce INSUFFICIENT_EVIDENCE; confirmation needs can produce WAIT_FOR_CONFIRMATION. PROTECT does not certify a stop is installed. EXIT_RECOMMENDED does not close P42: TM must receive and reconcile actual broker fills, then terminate or revise the relevant monitoring intent.

Illustrative record	Evidence / ownership to inspect
S1 → Advice-1 / MAINTAIN	TI: intact supported thesis; TM: current position snapshot. Both references remain in the A5 capture.
S2 → Advice-2 / PROTECT	TI: cited weakening/protection basis. TM: separate protection decision and observed broker-order status—not an inferred stop.
S3 → Advice-3 / EXIT_RECOMMENDED	TI: cited invalidation/horizon-risk reason. TM: new authorized exit intent. Broker: actual exit fill before closed exposure is recorded.

READ WITH A5 architecture §§3–7,11; Monitoring §§5,7–8

IV • CASE B / ILLUSTRATIVE FUTURE INTEGRATION

RELIANCE: positional equity without false A6

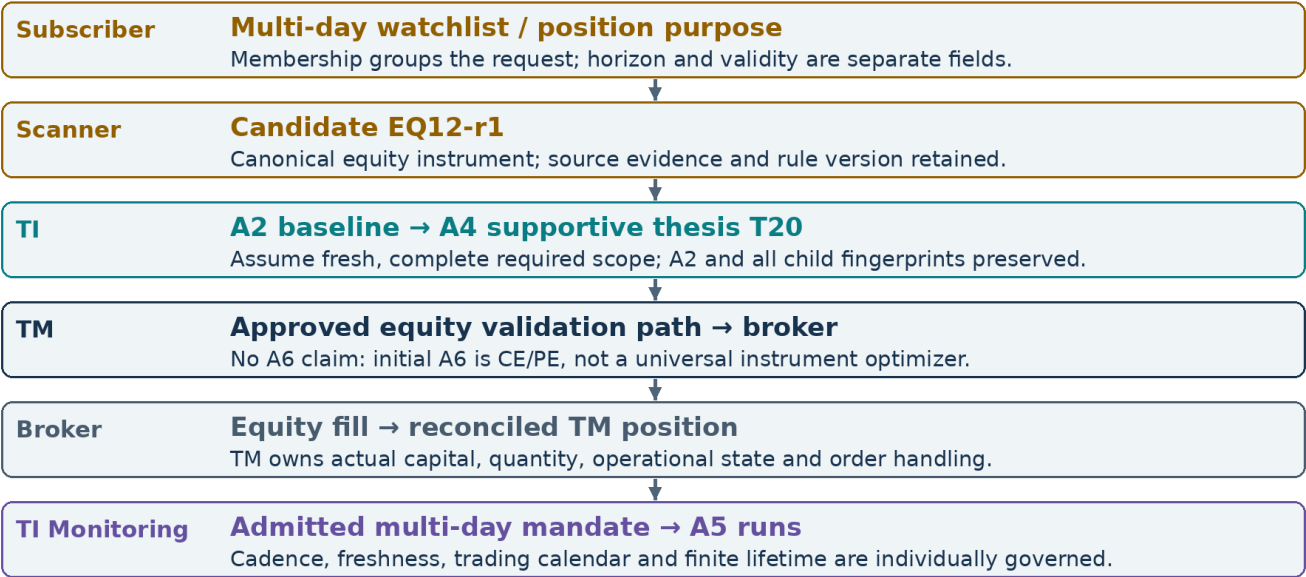


Figure 33. A valid equity path need not masquerade as an option-expression result.

Suppose the first A5 result is MAINTAIN and later evidence weakens the thesis. TI produces a new attributed recommendation, not a rewritten original thesis. An overnight gap or stale mark cannot be treated as fresh just because the mandate horizon spans several days. TM rechecks current broker truth and authority before responding.

READ WITH Ecosystem §§5–8; Monitoring clocks and calendar gate

IV • CASE C / ILLUSTRATIVE FUTURE INTEGRATION

HDFCBANK: adopting a manual broker position



Figure 34. The lineage starts with an external broker observation, not a fictional scanner event.

A later independently admitted TI evaluation may supply thesis evidence for a successor A5 run. That analytical work does not retroactively make the original manual trade TI-validated. TM’s adoption log records what was known and permitted at adoption time.

READ WITH Ecosystem §8; A5 insufficient-evidence and freshness rules

IV • CASE D / ILLUSTRATIVE CONFLICT

Scanner BUY, TI WAIT

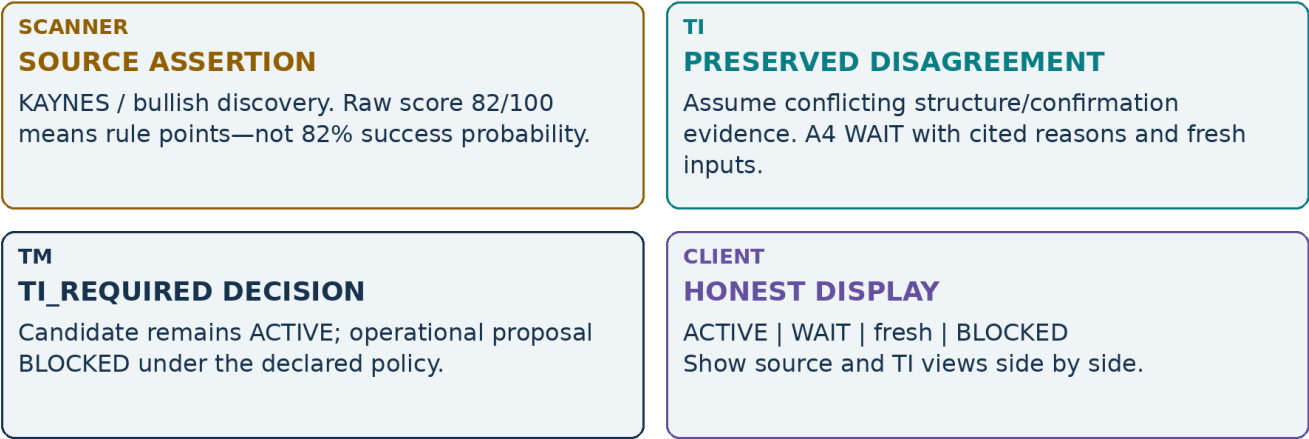


Figure 35. A disagreement is useful information, not a defect to tune away.

Do not average source score and TI confidence, call WAIT a provider outage, or rerun specialists until one produces SUPPORTIVE. A successor evaluation needs a bounded new trigger or changed admitted evidence and retains the prior disagreement. A2 NO_TRADE gates and declared blocking dispositions remain intact.

The future Signal Qualification layer must also preserve this acyclic discipline. Its output, if later implemented, names policy, scope and validity; it is not an unconditional “qualified to trade” badge.

Transparency is more valuable than making a live example look attractive.

READ WITH Ecosystem §§3,6–7; A2 benchmark and A4 preservation principles

IV • CASE E / ILLUSTRATIVE FAILURE SCENARIO

TI outage: stop entry, preserve operations



Figure 36. An outage changes availability, not the meaning of required scope or authority.

Recovery creates newly admitted evaluations and current TM reviews. It does not replay queued client approvals automatically. A prior unfavorable TI result stays recorded; optional missing-data handling cannot hide a known policy veto by treating it as unavailability.

READ WITH Ecosystem §§7,11; Monitoring failure isolation

IV • CASE F / ILLUSTRATIVE DEDUPLICATION

Two scanners, one symbol, distinct claims

Incoming event	Receiver treatment	Meaning preserved
A / C17 / r2 / digest H2 first receipt	Accept immutable candidate revision	Source A’s KAYNES assertion
A / C17 / r2 / digest H2 repeated delivery	Idempotent duplicate; retain receipt record	One candidate, not a second signal
A / C17 / r2 / digest H9 changed body	Conflict; quarantine/reject overwrite	Same revision cannot silently change meaning
A / C17 / r3 / digest H3 expected predecessor r2	Accept successor if valid and ordered	Invalidate old approval linkage; keep history
B / C88 / r1 / digest HB same KAYNES subject	Independent candidate retained	Different source, rule or purpose; not a duplicate

SCANNER
CANDIDATE IDENTITY

Namespace + candidate + revision + semantic digest
Receiver trace/receipt data has separate identity.

TI
EVIDENCE INDEPENDENCE

Two scanners quoting one source do not create two independent confirmations.

Figure 37. Idempotent delivery is not cross-source semantic collapse.

A shared instrument cluster may help navigation while preserving each source assertion. Canonical evidence admission separately decides rights, factual meaning and point-in-time availability. A changed nested expression is a candidate revision, not a mutable shared pointer. No deduplication step grants TM action authority.

READ WITH *Ecosystem §3; source-authority provenance/contradiction architecture*

IV • FUTURE INTELLIGENCE / FUTURE / TBD

Sector, qualification and forecasting

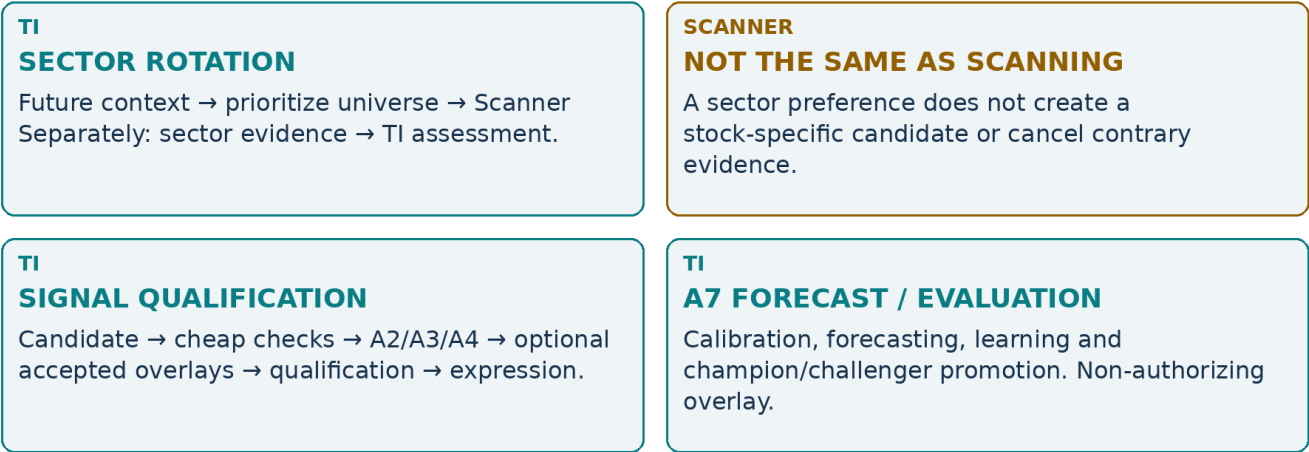


Figure 38. Separate future consumers, with no recursive search for a favorable answer.

Current A3.6 sector-context specialists are not a full Sector Rotation engine. Initial deterministic A6 does not require A7. Future calibrated evidence must be explicitly admitted and versioned; raw scanner confidence is not a substitute. Sector ordering relative to A7/A8 remains unresolved, and Signal Qualification follows its review unless explicitly reordered.

READ WITH Ecosystem §6; authoritative roadmap; TBD sector/qualification/forecasting notes

IV • EVOLUTION / CURRENT → FUTURE / TBD

A gated roadmap, not a wishful timeline

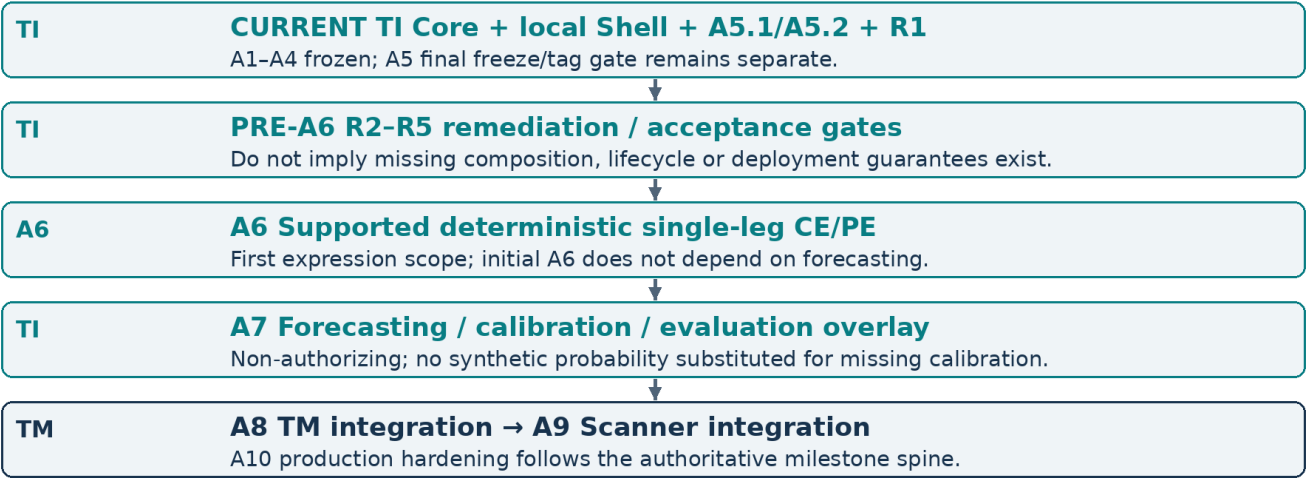


Figure 39. Normative milestone spine; this is not a delivery-date commitment.

Contract design may precede runtime integration. Monitoring runtime, Sheets/client integrations and a Web Cockpit each require separate admission, deployment and acceptance gates; this handbook does not assign them invented milestone numbers. Their dependency story is not permission to reorder the authoritative roadmap.

Sector Rotation’s placement relative to A7/A8 remains TBD; Signal Qualification follows that review unless explicitly reordered. The illustrative “scanner → monitoring → cockpit” journey explains dependencies, not a competing delivery sequence.

READ WITH TRADINGINTELLIGENCE_ROADMAP.md: current sequence; ecosystem §13

Boundaries to carry into implementation

Boundary still requiring acceptance	Do not infer
Source ingress authentication / published wire schemas	A payload source_id is authenticated; design field names are an implemented API
A6 eligibility, exposure sides and numeric policy	Every named stock has a supported option, or short-option authority exists
TM strategy, grants, action idempotency and reconciliation	TI support authorizes capital/quantity or a timeout means no fill
Fresh-input public binding / deployment / monitoring runtime	Captured replay acquires current facts or A5 hints start a daemon
Persistence, retention, cross-system export and R2-R5	One shared database or complete captured composition already exists
Sheets / cockpit authentication, session and UI technology	A shared screen shares authority, or credentials belong in a spreadsheet
Sector Rotation / qualification / calibrated A7 evidence	Research ideas are current services or a raw score is a forecast

Memorable rules

Scanners propose. TI advises. TM authorizes. Brokers execute. Watchlist is organization; mandate is semantics. Shared evidence is not shared analysis state. Deduplicate evidence, not intent. Monitoring is intelligence orchestration, not execution. Broker truth outranks cached assumptions. Replay explains; replay never executes.

This thesis is a human-readable companion. Normative architecture remains in the authoritative Markdown architecture documents.

The handbook changes no runtime contract, enum, recommendation policy, authority boundary or package/schema version. Its mock objects and screens are teaching devices, not a new specification. When a simplification needs precision, follow the source map on the next page.

REFERENCE / REFERENCE

Source map & compact glossary

Topic	Authoritative repository reference
Ecosystem owner / object / state model	docs/TIAF_TRADING_ECOSYSTEM_ARCHITECTURE.md
TI boundary and deployment	docs/TIAF_SYSTEM_ARCHITECTURE.md docs/TIAF_DEPLOYMENT_ARCHITECTURE.md
Mandates, clocks, sharing, scheduling	docs/TIAF_MONITORING_ARCHITECTURE.md
Replacement guarantees and required absence	docs/TIAF_PLUGGABILITY_ARCHITECTURE.md
Evidence meaning and contradiction	docs/ TIAF_SOURCE_AUTHORITY_PROVENANCE_CONTRADICTION_ARCHITECTURE.md
Challenge / arbitration	docs/ TIAF_A4_CHALLENGE_ARBITRATION_HIGHER_INTELLIGENCE_ARCHITECTURE.md
Single-position intelligence	docs/TIAF_A5_POSITION_INTELLIGENCE_ARCHITECTURE.md
Implementation truth, deferrals and order	docs/TIAF_CAPABILITY_MAP.md docs/TIAF_DEFERRAL_REGISTER.md docs/TRADINGINTELLIGENCE_ROADMAP.md
Narrative context	docs/TIAF_THESIS.md TI_Intelligence_Hierarchy_Thesis.docx TI_Monitoring_Architecture_Thesis.docx

Term	Read it as
PIT / as_of	Point-in-time availability / the explicit decision cutoff; not today’s knowledge.
Semantic fingerprint	Digest of declared meaning; not permission, freshness or an invented composition manifest.
OperationalPosition	TM’s reconciled managed lifecycle linked to broker truth, not a TI thesis.
MonitoringMandate	TI-admitted bounded intelligence work with separate subscriber intent.
CE / PE; A6 / A7	Call / put; future supported expression / calibrated forecasting-evaluation overlay.

READ WITH Version snapshot: 13 September 2026, Asia/Kolkata. Future idea caches are context only.